ISHAN CHAKRABORTY



QA Engineer & AI/ML Developer | Python Backend Developer

AI/ML Testing Specialist | RAG & Agentic AI Enthusiast | Full Stack Developer

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🔗 LinkedIn Profile | 💻 github.com/Ishan96Dev

🌐 Portfolio Website

PROFESSIONAL SUMMARY

Results-driven QA Engineer, AI/ML Developer & Python Backend Developer with 4+ years of experience specializing in building and testing AI/ML systems, REST API development, automation frameworks, and full-stack quality assurance. **Passionate about cutting-edge technologies — particularly RAG (Retrieval-Augmented Generation) systems, Agentic AI architectures, OpenAI/Google Gemini/Anthropic Claude API integration, and real-time financial data systems.** Hands-on expertise across all major AI providers: OpenAI (GPT, DALL-E, Whisper, TTS), Google Gemini (Pro, Flash, Embedding), and Anthropic Claude. Skilled in embedding models, RAG architectures using LangChain, LlamaIndex, CrewAI, Milvus, and ChromaDB. Proven expertise in testing Generative AI platforms, LLM models, RAG systems, and multi-agent AI workflows. Hands-on experience building Python backend APIs for real-time stock market data (FinTech), integrating OpenAI GPT-4/GPT-5 APIs, and developing MCP servers for AI developer tooling. Proficient in both frontend (Next.js, React.js, TypeScript) and backend (Python, FastAPI, Flask) development with comprehensive knowledge of API testing, containerization, and CI/CD pipelines. Skilled in developing robust test automation frameworks using Robot Framework, Selenium, Playwright, and Postman for API validation. Strong analytical capabilities in data-driven testing with experience in SQL, Supabase, data analytics, and performance testing tools. Committed to delivering software quality, reliability, and optimal performance in AI-powered applications, real-time financial data platforms, and enterprise systems.

CORE QA ENGINEERING CAPABILITIES

✨ AI/ML & Generative AI Testing

✓ LLM Model Testing & Validation
✓ RAG System Quality Assurance
✓ Agentic AI Workflow Testing
✓ Generative AI Chatbot Testing
✓ NLP Model Evaluation

Test Automation & Frameworks

✓ Robot Framework Automation
✓ Selenium WebDriver
✓ Playwright Automation
✓ pytest & Unit Testing
✓ Docker-based Test Execution

Full-Stack Testing

✓ Frontend: React.js, Node.js
✓ Backend: Python, REST APIs
✓ API Testing (Postman)
✓ Database & SQL Testing
✓ Integration Testing

Performance & Load Testing

✓ Locust Load Testing
✓ Apache JMeter
✓ Stress Testing
✓ Performance Metrics Analysis
✓ Scalability Testing

✓ Integration Testing

✓ Scalability Testing

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C#, C++, Java

AI/ML Technologies: Large Language Models (LLM), Natural Language Processing (NLP), RAG Systems, Agentic AI, Generative AI, OpenAI API (GPT, DALL-E, Whisper, TTS), Google Gemini API (Pro, Flash, Embedding), Anthropic Claude API, MCP Servers, LangChain, LlamaIndex, CrewAI, HuggingFace, Milvus, ChromaDB, Image Generation (DALL-E, Imagen), Text-to-Speech (TTS), Speech-to-Text (STT / Whisper), scikit-learn, TextBlob NLP, Machine Learning Model Testing

Frontend Technologies: React.js, Next.js, Node.js, TypeScript, JavaScript, HTML5/CSS3 Animations, Streamlit

Backend Technologies: Python, REST APIs, FastAPI, Flask, Node.js, Real-time Data APIs, Financial Data Systems (FinTech)

Testing Tools: Robot Framework, Selenium WebDriver, Playwright, pytest, Postman (API Testing), Locust (Load Testing), Apache JMeter

Containerization: Docker, Kubernetes, Docker Compose, Container-based Test Execution

Database & SQL: SQL, MySQL, PostgreSQL, Supabase, Database Testing, Query Optimization

Data Analytics: Data Visualization (Plotly, Apache Tableau), Data Analysis, Excel Analytics

Version Control & CI/CD: GitHub, GitLab, Jira, Jenkins, GitHub Actions

PROFESSIONAL EXPERIENCE

Sr. Executive (Technology) — June 2026 – Present

Metropolitan Stock Exchange of India (MSE)

• **Full-Stack Development:** Serving as a full-stack developer building and maintaining exchange technology platforms for India's emerging stock exchange.
• **Backend Services:** Developing backend services and RESTful APIs using Node.js and Python to support trading and market data systems.
• **Frontend Development:** Building responsive frontend interfaces and robust backend systems with Next.js; maintaining the official MSE website.
• **Internal Projects:** Contributing to internal MSE development app projects, delivering features across both frontend and backend layers.
• **Source Control:** Managing source code workflows via GitHub and GitLab with structured branching strategies and code reviews.
• **Linux Administration:** Administering Linux servers and handling MS SQL database operations to support trading and market data systems.

Freelance Python Backend Developer

April 2026 – Present

Metropolitan Stock Exchange of India (MSEI) | Remote

• **FastAPI Backend Service:** Built production FastAPI (Python) backend for Metropolitan Stock Exchange of India (MSE) delivering real-time SX40 equity index and USDINR/EURINR/GBPINR/JPYINR currency derivatives ticker data via REST API; engineered dual-mode operation — live TCP exchange feed (FEED_HOST/FEED_PORT) with 3s auto-export and TOUCHLINE.TXT file-fallback with 10s auto-export, ensuring zero-downtime continuity

• **LED Display Board Integration:** Developed dedicated LED ticker routes (/led, /led-single) with pixel-pitch-aware geometry-tuned typography (bold 64px+) calibrated for physical 28m × 160mm P1.8 LED wall panels (~15,556×89 px effective resolution); replaced CSS keyframe approach with requestAnimationFrame motion engine and per-frame movement clamping to eliminate LED scan/refresh artifacts

• **Animated Ticker UI & Market-Close Intelligence:** Built dual-row (/) and single-row (/single) scrolling HTML ticker pages with viewport-aware JS animation, sticky-direction logic preserving trend indicators on unchanged LTP, market-close auto-detection at 3:30 PM IST with IST-formatted inline closed-market notices, Yesterday Market prefix after midnight IST, and automatic page-reload on open/closed state transitions

• **JSON REST API & Next.js Integration:** Exposed /api/ticker-ui and /exports/ticker-ui.json endpoints returning UI-ready structured JSON consumed by client Next.js website (iframe blocked by X-Frame-Options: DENY); hardened API with SecurityHeadersMiddleware, SlowAPI rate limiting, CORS (ALLOWED_ORIGINS), and API key authentication; warm-start snapshot restore ensures continuity on service restart

• **Testing & Documentation:** Authored 35-test Python unittest suite covering export schema, sticky-direction logic, market-close scheduling, LED geometry, and warm-start behavior — all passing; delivered Swagger UI (/docs), full README.md, and docs/vm-nextjs-integration.md with cross-platform launchers (start.ps1 / start.sh) for Windows and Linux/RHEL VM deployment

QA Engineer

January 2024 – December 2025

Katonic.ai | Remote

• **Automation Framework:** Executed comprehensive automation testing using Robot Framework with Docker containerization for scalable test execution across multiple VM/Container environments

• **CI/CD Integration:** Developed custom mailing system for automated test report delivery in GitHub CI/CD pipelines using Robot Framework, HTML, and Python for seamless communication of test results

• **Cross-Browser Testing:** Developed and maintained automated test suites using Selenium WebDriver and Playwright for comprehensive UI automation

• **API Testing:** Performed extensive API testing using Postman, validating REST endpoints, request/response payloads, and authentication workflows

• **Performance Testing:** Conducted load and performance testing using Locust and Apache JMeter to evaluate system scalability and identify bottlenecks

• **LLM Testing:** Executed specialized testing of Large Language Models and Generative AI systems to ensure accuracy, reliability, and response quality

• **RAG Systems:** Validated RAG (Retrieval-Augmented Generation) system performance, testing document retrieval accuracy and contextual response generation

• **Agentic AI:** Tested Agentic AI workflows, ensuring proper task execution, decision-making logic, and multi-agent coordination

• **Test Management:** Developed and executed test cases using Excel, leveraging AI tools and data analytics for test coverage

optimization
• **Defect Tracking:** Efficiently logged, tracked, and managed defects using GitHub Issues and Jira for streamlined bug resolution
• **Database Testing:** Performed database testing and SQL query validation to ensure data integrity and backend functionality
• **Full-Stack Collaboration:** Collaborated with frontend (React.js, Node.js) and backend (Python) development teams for end-to-end quality assurance

Data Scientist **September 2021 – January 2024**

Katonic.ai | Remote

• Led predictive analytics project on bank churn incident prediction for PNC Bank, delivering actionable insights through machine learning models
• Developed and deployed production-ready applications for Katonic Marketplace using Streamlit framework
• Created Sentiment Analyzer application using NLP techniques for text classification (positive, negative, neutral sentiment detection)
• Built Mask Detection application leveraging computer vision and deep learning models
• Tested and contributed to Katonic MLOps & GENAI Platform, ensuring quality across AI/ML pipelines
• Developed and deployed Generative AI chatbots using industry-specific data from banking and financial sectors
• Created interactive data visualizations and dashboards using Plotly for complex data analysis

Test Engineer Intern **January 2020 – March 2020**

Sectona | Mumbai

• Conducted thorough testing of security-related user interfaces for enterprise security software

• Successfully integrated and tested UI compatibility with both Linux and Windows environments

• Performed manual testing and documented test cases for security feature validation

🚀 PERSONAL PROJECTS

🤖 Document Knowledge Retrieval (RAG System)

A multi-agent RAG (Retrieval-Augmented Generation) system powered by CrewAI, Milvus, and OpenAI. Upload documents and ask questions using natural language - AI agents will find relevant information and synthesize comprehensive answers.

🔗 View on GitHub | 🚀 Live Demo

🚀 Digital Portfolio App

A modern, interactive digital portfolio built with Streamlit featuring dual themes and AI/ML-inspired effects. Comprehensive web application designed to showcase professional profiles, skills, and achievements with modern CSS animations.

🔗 View on GitHub

📊 EA Stock Streamlit Dashboard

Interactive Streamlit dashboard combining EA stock data (price, dividends, stock splits) with user behavior analysis. Features dynamic charts, filters, and insights into historical stock performance and user engagement metrics.

🔗 View on GitHub

🤖 Robot Framework GitHub Action

GitHub Action for running Robot Framework tests using Docker containerization (ppodgorsek/robot-framework base image). Enables automated CI/CD testing workflows for continuous integration pipelines.

🔗 View on GitHub

🧠 Machine Learning Projects

• Sentiment Analyzer App: NLP-based application that analyzes sentiment in text (positive, somewhat-positive, negative, somewhat-negative, or neutral). Uses natural language processing techniques to detect sentiment in social data.
• Face Mask Detector Using OpenCV & Python: Computer vision solution addressing COVID-19 safety compliance by automatically detecting face mask usage. Built with Python, OpenCV, and TensorFlow.

🔗 View on GitHub

🤖 MCP OpenAI Image Generator

Python-based Model Context Protocol (MCP) server integrating OpenAI image generation APIs with Claude, Cursor, VS Code, and GitHub Copilot. Features rate limiting, error handling, and server deployment on Render. Enables seamless AI image generation within developer workflows.

🔗 View on GitHub

📄 DocForge — Intelligent Web-to-PDF Converter

TypeScript + React + FastAPI powered web application for smart web crawling and sitemap detection that converts live web documentation into structured PDF files. Features intelligent page discovery and batch PDF generation.

🔗 View on GitHub | 🚀 Live Demo

🌐 Ishan Portfolio — Developer Portfolio

Official developer portfolio deployed on Vercel, showcasing AI, Automation, and Web Development projects. Built with JavaScript featuring responsive design and modern UI/UX patterns.

🔗 View on GitHub | 🚀 Live Site

🔍 SentimentScope — NLP Sentiment Analysis

Real-time NLP sentiment analysis application built with Python and Streamlit using TextBlob for emotion detection with confidence scoring. Provides instant sentiment classification (positive/negative/neutral) with visual confidence metrics.

🔗 View on GitHub

💰 YesBill — AI-Powered Billing Tracker

Full-stack billing management app built with React + FastAPI + Supabase + GPT-4. Features AI-powered daily service billing, YES/NO delivery marking, automated bill calculation, PDF invoice export, and AI-generated spending summaries.

🔗 View on GitHub | 🚀 Live App

🔗 HeliosConnect — LinkedIn AI Networking Agent

AI-powered LinkedIn automation agent built with Python featuring semantic prospect scoring, GPT-4 message generation, and rate-aware connection automation. Automates intelligent outreach with personalized AI-generated messages.

🔗 View on GitHub

🧠 AI Sentiment Analyzer — GPT-4/5 Powered

Advanced AI sentiment analysis system integrating OpenAI GPT-5/GPT-4 with real-time validation, reasoning transparency, and 4-type sentiment classification. Features confidence metrics and professional Streamlit UI for enterprise-grade text analysis.

🔗 View on GitHub

🎓 EDUCATION

Bachelor of Engineering (Electronics & Communication)

RKDF University, Bhopal | 2019

College Internship: ZCODE TECHNOLOGY - C++, Java, Python, and IC Programming

Higher Secondary Certificate

Lords Universal Junior College, Mumbai | 2015

Specialization: Electronics 1 and Electronics 2

ICSE Secondary Education

Prime Academy ICSE & International, Mumbai | 2012

Specialization: Technical Drawing Application

CERTIFICATIONS

Python Programming
Udemy

Docker & Containerization
Udemy

Google Analytics
Basic & Advanced

ACTIVITIES & INTERESTS

⚽ Football • ♟️ Chess • 🎮 Game Development (Unity & C#) • 🔬 Testing Emerging Technologies • 📸 Photography • 🎯 Video Gaming • 🎬 Video Editing • 🍳 Cooking

CANDIDATE DECLARATION

I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief. I understand that any false or misleading information may result in the rejection of my application or termination of employment if discovered at a later date.

Date: 11/11/2026
Place: Mumbai, Maharashtra, India

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(Ishan Chakraborty)

Candidate Signature